CoRoT-2b

R-0178

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_180822` · created 2026-07-16T18:53:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458352.7556 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.009 +/- 0.066
Transit depth (Rp/Rs)^2	0.0084 +/- 0.12 [%]
Orbital Inclination (inc)	83.47 +/- 2.07
Airmass coefficient 1 (a1)	11.14 +/- 6.76
Airmass coefficient 2 (a2)	-1.81 +/- 0.9
Scatter in the residuals of the lightcurve fit is	71.29 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	6.5
Transit Duration (day)	0.054 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.009 ± 0.066 vs 0.1667 (deviation 2.39σ)
Duration fit vs published	0.054 d vs 0.095 d (deviation 1.64σ)
Mid-time O–C	4.6 min
Depth SNR	0.1
Residual scatter	71.29 %

Light curve

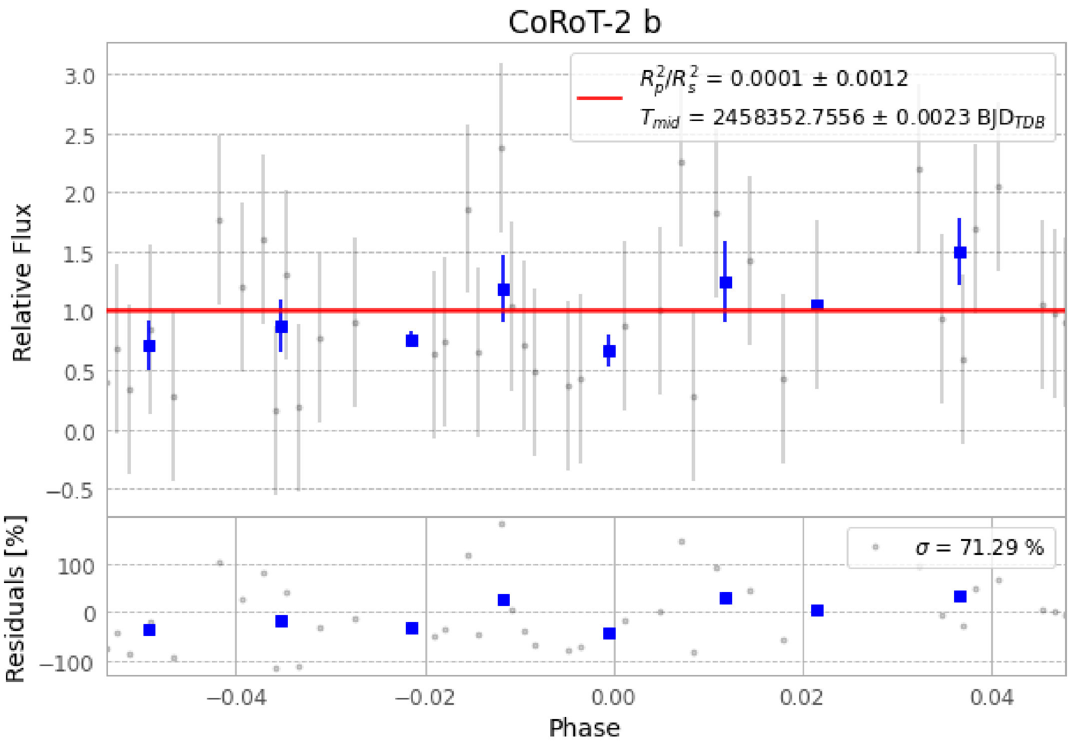

Figure 1 — FinalLightCurve_CoRoT-2 b_2018-08-21.png (SHA-256 cdc6862c21172048…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [447, 188], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 28aa6fdb5d9591e7…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca`

Data provenance

87 FITS frames (57 MB), `CoRoT-2180822034212.FITS` → `CoRoT-2180822080012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180822.log (c28ed43b0bb8…), FinalLightCurve_CoRoT-2 b_2018-08-21.png (cdc6862c2117…), FinalLightCurve_CoRoT-2 b_2018-08-21.csv (a2ef50e998d0…)

Compute resources

Wall time 235.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:53Z.